

Question	Answer	Marks	Guidance
6(a)(i)	time constant = RC	C1	
	resistance = $2.4 / (47 \times 10^{-6})$ $= 5.1 \times 10^4 \Omega$ $= 51 \text{ k}\Omega$	A1	Correct to at least 2 significant figures. AFC. (51.1)
6(a)(ii)	discharge time for each cycle = 2.5 s	C1	
	$V_{\text{MIN}} = 1.50 \exp(-2.5/2.4) = 0.53 \text{ V}$	A1	Full substitution and answer needed. (0.529) Unit not needed but any unit given must be correct.
6(b)	straight horizontal lines at $V_C = 1.50 \text{ V}$ from $t = 0$ to 2.5 s and $t = 5.0$ to 7.5 s	B1	Allow tolerance $\pm \frac{1}{2}$ small square vertically.
	smooth discharge curve of continuously decreasing negative gradient from $t = 2.5 \text{ s}$ to 5.0 s and $t = 7.5 \text{ s}$ to 10.0 s	B1	Do not allow if line becomes horizontal. <u>Special Case:</u> Allow one of the first two B1 marks if the phase is inverted but all otherwise correct.
	discharge curves show minimum voltage at $V_C = 0.53 \text{ V}$	B1	Allow tolerance $\pm \frac{1}{2}$ small square vertically. Allow any shape containing at least two minima with all minima shown at 0.53 V within tolerance and never below 0.53 V.
6(c)	smaller time spent discharging (before re-charging)	B1	
	minimum p.d. is greater (than 0.53 V)	B1	